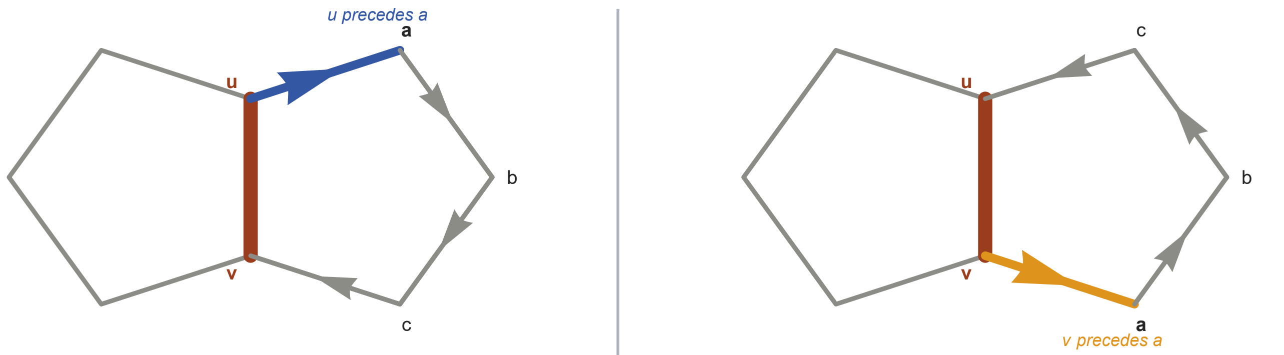
Anatomy of the pentagon mesh — action, object, invariant

1 · INPUT — the gluing ACTION: one letter per joint, an element of Z2 = Aut(shared edge)

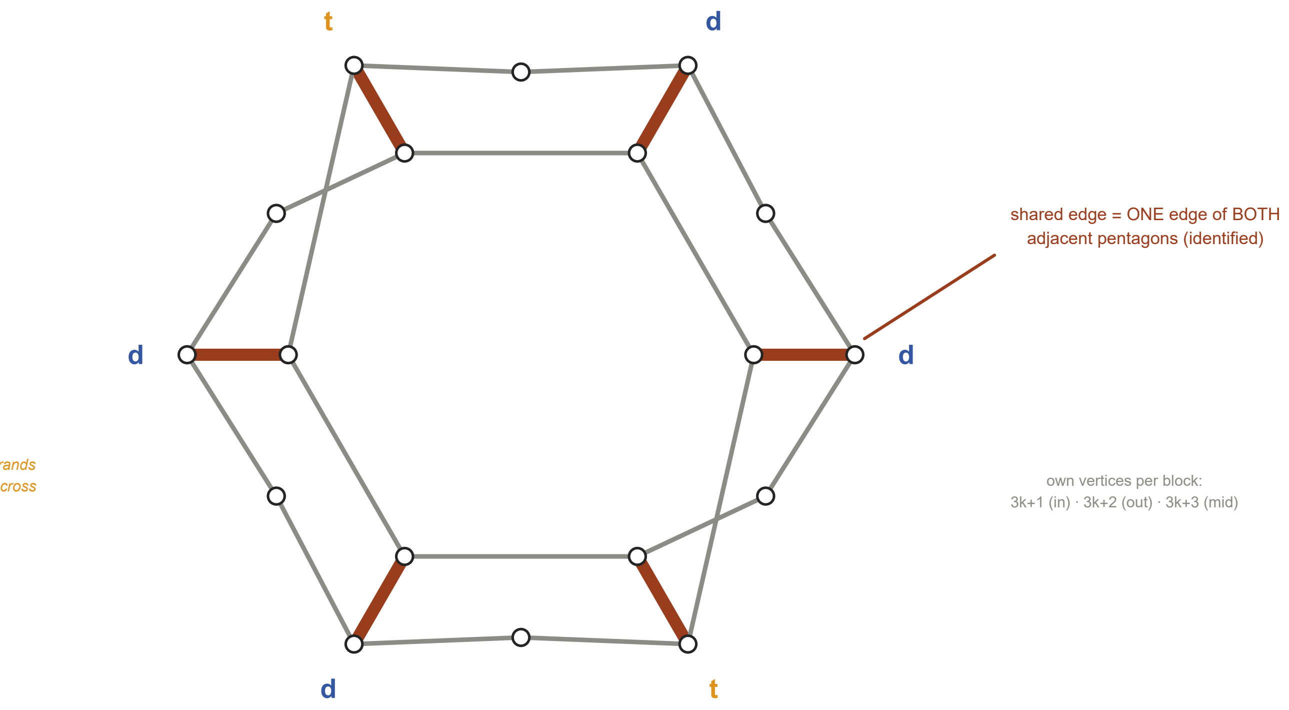


d — orientation kept (rotation)

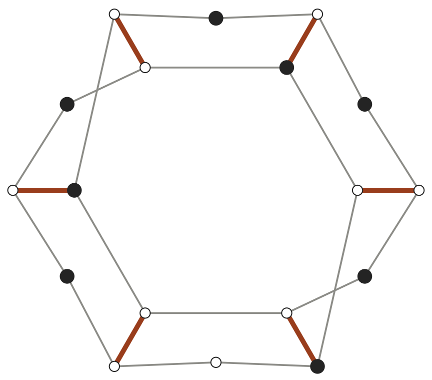
t — orientation flipped (reflection)

↓ *gluing = IDENTIFICATION (quotient): the coral edge is one edge of both blocks — pentagons overlap, they are not wired together*

2 · OBJECT — W = (ddt)^2 builds ONE exclusivity graph G_W: 3L = 18 vertices, 4L = 24 edges

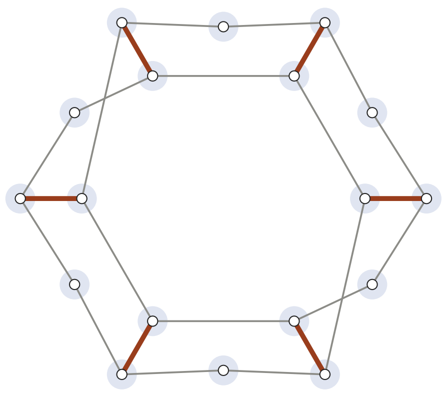


↓ *invariants live on the WHOLE composite graph — α composes letter-by-letter (transfer DP), ϑ does not (global SDP only)*



α(G_W) = 8 → 4/3 per block

classical bound — reached from the letters by the
3-state max-plus transfer DP: trace(T_d T_d T_t T_d T_d T_t) = 8
letter-local, exact (filled = a maximum independent set, live)



ϑ(G_W) = 8.34704 → 1.40323 per block (L → ∞)

quantum bound — ONE semidefinite program over all 18
vertices jointly; the optimal witness is delocalized (halo);
no per-letter factorization exists in the pipeline

3 · OUTPUT (continued) — gap density ϑ/L − α/L: a function of the WHOLE word, not of any letter